

# ATmega32A Module

## ADC (Analog-to-Digital Converter)

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### 1 Theory

The world outside the microcontroller is *analog*: a voltage can have any value, for example 1.37 V. The CPU can only work with numbers. An ADC (*Analog-to-Digital Converter*) measures a voltage and gives you a number.

**Resolution and reference.** The ATmega32A has a **10-bit** ADC. So the result is a number from 0 to  $2^{10} - 1 = 1023$ . The *reference voltage*  $V_{\text{REF}}$  is the largest voltage the ADC can measure. On our board  $V_{\text{REF}} = 5 \text{ V}$ . One step of the result (1 LSB) is

$$1 \text{ LSB} = \frac{V_{\text{REF}}}{1024} = \frac{5 \text{ V}}{1024} \approx 4.9 \text{ mV}$$

and the result for an input voltage  $V_{\text{IN}}$  is

$$\text{value} = \frac{V_{\text{IN}} \cdot 1024}{V_{\text{REF}}} \quad V_{\text{IN}} = \frac{\text{value} \cdot V_{\text{REF}}}{1024}$$

| Input voltage | 10-bit value | 8-bit value |
|---------------|--------------|-------------|
| 0 V           | 0            | 0           |
| 1.25 V        | 256          | 64          |
| 2.5 V         | 512          | 128         |
| 3.75 V        | 768          | 192         |
| 5 V           | 1023         | 255         |

If 8 bits are enough (for example for 8 LEDs or for PWM), we use only the upper 8 bits of the result: 0 to 255.

**Channels.** There is only one ADC, but it has a multiplexer with 8 inputs: **ADC0–ADC7** on the pins **PA0–PA7**. You choose the channel for each measurement.

**Conversion time.** The ADC needs its own clock between 50 and 200 kHz for the full 10-bit accuracy. It is made from the CPU clock with a prescaler. One conversion takes 13 ADC clock cycles: at 125 kHz this is about  $104 \mu\text{s}$ .

### 2 How to use the driver

The driver gives you one object: `ADC`. First you switch the ADC on one time with `ADC.enable()`. This does not measure anything yet. Then you call `ADC.read()` every time you need a new value: each call makes one measurement. When you do not need the ADC any more, `ADC.disable()` switches it off again, so it uses no power.

## Functions

| Function                                            | What it does                                                                                                                                                                                                           |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ADC.enable()</code>                           | Switch the ADC on (bit <code>ADEN</code> = <i>ADC Enable</i> ). It does not measure. Reference = <code>AREF</code> (5 V on our board). The driver chooses the ADC clock from <code>F_CPU</code> (8 MHz / 64 = 125 kHz) |
| <code>ADC.disable()</code>                          | Switch the ADC off ( <code>ADEN</code> = 0). It then uses no power. While it is off, every read returns 0                                                                                                              |
| <code>ADC.read(channel)</code>                      | Measure the voltage on <i>PAchannel</i> (channel 0–7) and return 0–1023. It waits until the conversion is finished (about 104 $\mu$ s)                                                                                 |
| <code>ADC.read8(channel)</code>                     | The same, but only the upper 8 bits: 0–255. Good for LEDs and PWM                                                                                                                                                      |
| <code>ADC.readMillivolts(channel)</code>            | The same, in millivolts: 0–4995 mV                                                                                                                                                                                     |
| <i>With interrupts (see the interrupt lecture):</i> |                                                                                                                                                                                                                        |
| <code>ADC.start(channel)</code>                     | Start one conversion and return at once (no waiting)                                                                                                                                                                   |
| <code>ADC.onComplete(function)</code>               | Call <code>function(value)</code> when a conversion is finished (value 0–1023)                                                                                                                                         |

**The pin.** `read()` makes the pin an *input* and switches its pull-up resistor *off*. A pull-up would change the voltage that you want to measure. So you do not need `GPIO.setDirection()` for an ADC pin.

**Reference voltage on our board.** On the course board the pins `VCC`, `AVCC` and `AREF` are all connected to +5 V. For this reason the driver always uses `AREF` as the reference. The internal 2.56 V reference must not be used on this board: it would be connected to the 5 V on the `AREF` pin.

## Driver functions and registers

In the lecture slides we set the registers by hand. Each driver function does these steps for you:

| Driver function                    | Registers                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ADC.enable()</code>          | <code>ADMUX = 0;</code> ( <code>REFS1:0</code> = 00: <code>AREF</code> ; <code>ADLAR</code> = 0)<br><code>ADCSRA = (1&lt;&lt;ADEN)   (1&lt;&lt;ADPS2)   (1&lt;&lt;ADPS1);</code> (prescaler 64 at 8 MHz)                                                                                                                                       |
| <code>ADC.disable()</code>         | <code>ADCSRA &amp;= ~(1&lt;&lt;ADEN);</code>                                                                                                                                                                                                                                                                                                   |
| <code>ADC.read(0)</code>           | <code>DDRA &amp;= ~(1&lt;&lt;PA0); PORTA &amp;= ~(1&lt;&lt;PA0);</code> (input, no pull-up)<br><code>ADMUX = (ADMUX &amp; 0xE0)   0;</code> ( <code>MUX4:0</code> = channel)<br><code>ADCSRA  = (1&lt;&lt;ADSC); while (ADCSRA &amp; (1&lt;&lt;ADSC));</code><br><code>return ADCW;</code> ( <code>ADCL</code> first, then <code>ADCH</code> ) |
| <code>ADC.read8(0)</code>          | <code>ADC.read(0) &gt;&gt; 2</code> (the same as <code>ADLAR</code> = 1 and reading only <code>ADCH</code> )                                                                                                                                                                                                                                   |
| <code>ADC.readMillivolts(0)</code> | <code>ADC.read(0) * 5000 / 1024</code>                                                                                                                                                                                                                                                                                                         |

### 3 Example: a small voltmeter

Board hardware.

- **Potentiometer** R13 (10 k $\Omega$ ) on PA0 = **ADC0**. Turn it to change the voltage on PA0 from 0 to 5 V.
- **8 LEDs (D0–D7) on PORTB**, active-low: LOW = ON. LED **D3** is on PB3 = 0C0, the PWM output of Timer0.
- **USB-UART** (FT232) on PD0/PD1: open a serial terminal with 9600 baud, 8N1 (see the UART handout).

**Build and upload.** `make adc`. Open the terminal and turn the potentiometer. You see for example ADC = 759 (3706 mV).

```
/*
 * ADC Example -- a small voltmeter: read the potentiometer and send the
 * result to the PC two times per second.
 * Build & flash: make adc
 *
 * Open a serial terminal on the PC: 9600 baud, 8N1. Turn the potentiometer
 * and watch the numbers, for example:
 *   ADC = 759 (3706 mV)
 *
 * The potentiometer of the board is on PA0 (ADC channel 0).
 */

#define F_CPU 8000000UL
#include "../uart/uart.hpp"
#include "adc.hpp"

int main() {
    UART.begin(9600);
    ADC.enable(); // reference AREF = 5 V, ADC clock 125 kHz

    while (true) {
        uint16_t value = ADC.read(0);           // 0..1023
        uint16_t mv = ADC.readMillivolts(0);    // 0..4995 mV

        UART.print("ADC = ");
        UART.print(value);
        UART.print(" (");
        UART.print(mv);
        UART.println(" mV)");

        _delay_ms(500);
    }
}
```

## 4 Laboratory Tasks

For each task, make a new .cpp file in the main project folder (for example `Task1.cpp`). Start the file with `#define F_CPU 8000000UL`. Then include the drivers you need: `drivers/adc/adc.hpp`, `drivers/gpio/gpio.hpp`, `drivers/timer/timer.hpp`, `drivers/uart/uart.hpp`. Upload it with `make SRC=Task1.cpp`.

### Task 1: Potentiometer on the LEDs (25 min)

Read the potentiometer (channel 0) and show the value on the 8 LEDs on `PORTB`. Use `ADC.read8(0)`, because there are only 8 LEDs. The LEDs are active-low, so write the inverted value: `GPIO.write(PORTB, ~value)`.

Turn the potentiometer from one end to the other. Which LEDs are on at the two ends? Which value do you read in the middle?

### Task 2: Potentiometer Sets the LED Brightness (20 min)

Use the potentiometer to set the brightness of LED **D3** (`PB3 = 0C0`):

1. Start `Timer0` in `TIMER_FAST_PWM` mode with `PWM_INVERTING` (the LED is active-low), as in the timer lesson.
2. In the loop, read the potentiometer with `ADC.read8(0)` and write the value with `Timer0.setCompare()`.

Now the potentiometer works like a light dimmer.

### Assignment: Find the Cross Voltage

The pin `PA0` can be read in two ways: as an *analog* value with the ADC, and as a *digital* value (0 or 1) with `GPIO.read(PINA, PA0)`. At some input voltage the digital value changes from 0 to 1. Find this *cross voltage*.

1. In the loop, read both values of `PA0`: `ADC.readMillivolts(0)` and `GPIO.read(PINA, PA0)`.
2. Send both to the PC with the UART (Lesson 6), for example `3706 mV -> 1`, about 10 times per second.
3. Turn the potentiometer *slowly* from 0 V up to 5 V. At which voltage does the digital value change from 0 to 1?
4. Now turn it slowly back down. At which voltage does it change from 1 to 0? Is it the same voltage? Compare your results with the input voltages  $V_{IL}$  and  $V_{IH}$  in the datasheet (*Electrical Characteristics*).